ISA AND CORE2350B (RP2350B)

SAMPLE RAM: on the module

The Core2350B2 has 8 MB of PSRAM on the RP2350B's QMI (CS1 = GPIO47, memory mapped). It replaces v2.0's U8 (APS6404L on GPIO0–3, PIO SPI); GPIO0–3 are now free.

I2S DAC

POWER

IRQ/DMA SETTING

MISC LEVEL SHIFT/BUFFER

AEN IO MASKING

Disables IOR and IOW signals when other DMA in system happens

ADDR/DATA MUX/DEMUX

RESET CIRCUIT

Latches BUSOE high after ADS is toggled for the first time, sets low on RESET

I/OCHRDY OPEN DRAIN

MOUNTING

74AHCxx LOGIC PWR

DACK INVERT

Saves PIO instructions due to JMP limitations

DECOUPLING CAPS

MIDI/DEBUG OUT

U6 74AHCT126 on +5V: OE high pulls the output low, OE low leaves it open (replaces the 74LVC2G06). Spare gates: A and OE tied low.

BOOT ROM (experimental)

Boot ROM

File: bootrom.kicad_sch

SST39SF010A in a PLCC-32 socket + 74HCT138: uses the global nets A0–A19, D0–D7, ~SMEMR and ~ROM_EN (GPIO2).

RADIO MODULE 2: WiFi / Bluetooth (CYW43439)

gSPI as on the Pico W: data in, data out (470R) and the IRQ (10k) share one GPIO. GPIO40–45 are the RP2350B pins that are not 5 V tolerant: fine here, all 3.3 V. Antenna end on the bracket edge; RF keep-out on the board (see README).

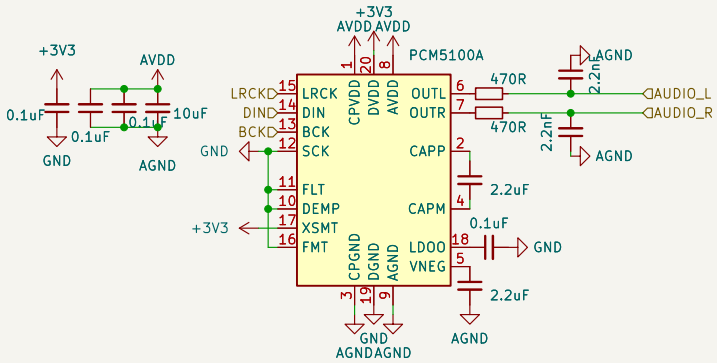
USB-C PROGRAM PORT

5.1k Rd on CC1/CC2: device only, never sources VBUS. D+/D- shared with J5 (USB host) as on v2.0.

v2.0 ISA logic and GPIO map; TSSOP/SOIC-only chips as SMD
Through-hole PicoGUS 2.0 on a Waveshare Core2350B (RP2350B) module

Sheet: /	
File: PicoGUS-THT.kicad_sch	
Title: PicoGUS v2.1 THT	
Size: A3	Date: 2026-10-29
KiCad E.D.A. 10.0.7	Id: 1/4

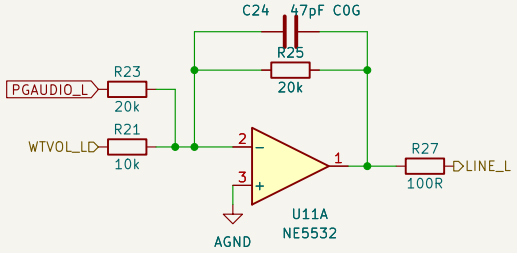
Rev: v2.1

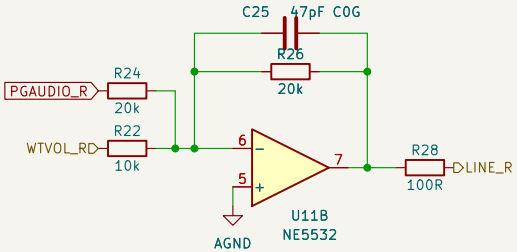


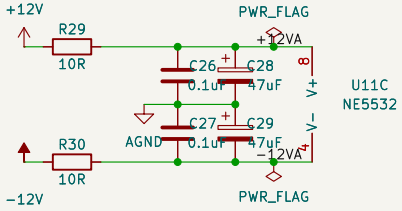
Sheet: /Audio/		
File: audio.kicad_sch		
Title: PicoGUS v2.1 THT – PCM5100A DAC		
Size: A4	Date: 2026-10-29	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 2/4

LINE-OUT MIXER

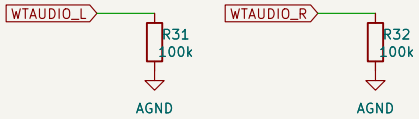
Inverting summing amplifier from THT v1.3: Waveblaster (after U13 M62429 volume) x2 = +6 dB through R21/R22 10k.
PCM5100A x1 through R23/R24 20k, feedback 20k || 47 pF (pole ~170 kHz). U11 runs from ISA +/-12V through 10R / 47uF.
The DAC output is ground-centred, so the PCM path is DC coupled (no electrolytic); the M62429 output sits at about
VCC/2, so C41/C42 (film) couple it. Both channels invert, so left and right keep their relative phase.







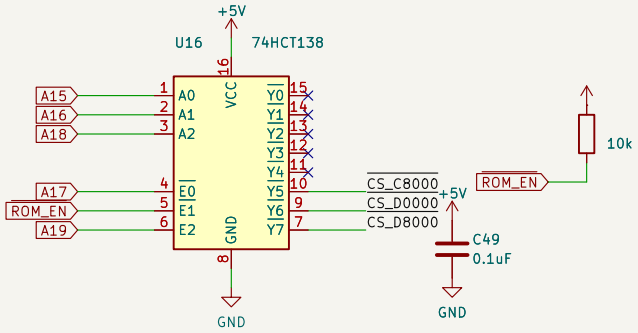
WT input pull-downs (THT v1.3 R14/R15)

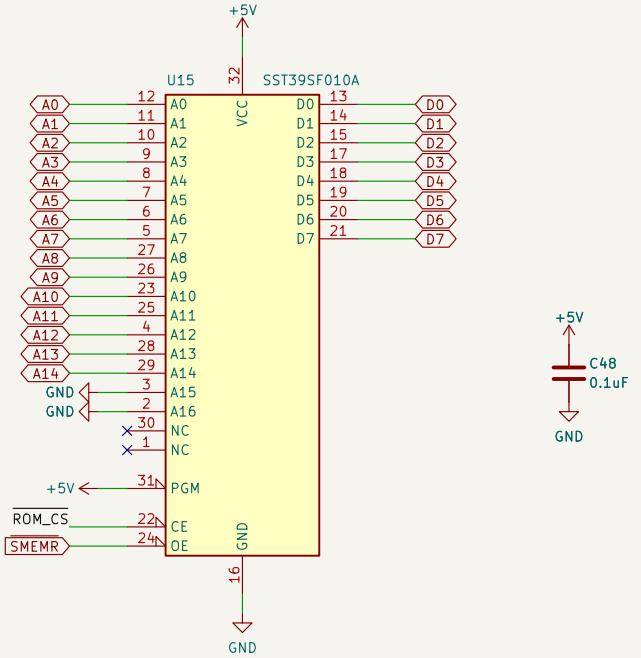


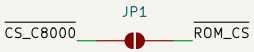
Sheet: /Mixer/		
File: mixer.kicad_sch		
Title: PicoGUS v2.1 THT – line-out mixer		
Size: A4	Date: 2026-10-29	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 3/4

BOOT ROM: 32 KB option ROM on the ISA memory bus (experimental)

U16 decodes SA19..SA15: outputs 5, 6 and 7 are low in C8000h–CFFFFh, D0000h–D7FFFh and D8000h–DFFFFh while ~ROM_EN (GPIO2) is low.
R35 keeps the ROM off until the firmware enables it. The closed jumper picks the block: its output is the flash CE; OE is ~SMEMR,
so the flash drives the ISA data lines only during a memory read in its window.











WE is tied high: program the flash in a programmer (it is in a socket).

Base address: close exactly one of JP1 (C8000h), JP2 (D0000h, default: bridged by a track, cut it to change) and JP3 (D8000h).

Sheet: /Boot ROM/		
File: bootrom.kicad_sch		
Title: PicoGUS v2.1 THT – boot ROM (experimental)		
Size: A4	Date: 2026-11-11	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 4/4